

TIANQI KOU

The Pennsylvania State University

tfk5237@psu.edu

www.koutianqi.info

BACKGROUND

I am a philosopher of science trained in mathematics, computer science, and history and philosophy of science and technology. My research explains why AI hype persists in science and engineering through the lens of epistemic values and political-economic norms, and addresses it through conceptual and practical solutions. My teaching experience spans scientific, technical, and humanities subjects, with an emphasis on human-centered science and engineering.

RESEARCH KEYWORDS

History and Philosophy of Science and Technology, Scientific Replicability, Machine Learning Replicability, Computing Ethics, Science and Technology Studies, Science Policy, Technology Policy

EDUCATION

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| 2023–27 | Ph.D., The Pennsylvania State University
Information Science (Advisor: Dana Calacci)
Committee: Andrea Miller, Leif Hancox-Li, Cindy Lin
Dissertation: <i>Toward a Feminist Conception of Replicability for Machine Learning Research</i> |
| 2020–23 | M.S., The Pennsylvania State University
Information Science (Advisors: Daniel Susser, Fred Fonseca) |
| 2017–19 | M.S., Dean’s Fellowship, Fordham University
Computer Science (Advisors: Yijun Zhao, Chaitra Nagaraja) |
| 2013–17 | B.S., with high honors, Harbin Institute of Technology, Monash University
Economics |

APPOINTMENTS & AFFILIATIONS

Working Futures Network

2024–	PhD Student Researcher Director: Dana Calacci
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LOGIC(S)

2024–25	Liberal Tech Scholar Fact Checker
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Critical Technocultures Lab, Georgia Institute of Technology

2023–25	Visiting Scholar
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Director: Cindy Lin

Center for Socially Responsible Artificial Intelligence, Penn State University
2022– Student Affiliate

Apteo
2018–20 Machine Learning Engineer

AWARDS & FELLOWSHIPS

2026 Harold F. Martin Graduate Assistant Outstanding Teaching Award, Penn State
2025 Graduate Student International Travel Award, Penn State Graduate School
2025 Dean’s Travel Award
2025 Sloan Foundation Postdoctoral Fellow Finalist
2024 Liberatory Tech Scholar Fellowship, LOGIC(S)
2024 FAccT Student Travel Award
2023 Dean’s Travel Award, Penn State IST
2022 Microsoft PhD Research Fellowship Nomination, Penn State IST
2019 Graduate Academic Excellence Award, Fordham University
2017–19 Dean’s Fellowship, Fordham University
2016–17 Chinese Scholarship Council First Class Scholarship
2013–17 National People’s Scholarship

PUBLICATIONS

Note: In computer and information science, papers published in major conference proceedings are double-anonymously peer-reviewed and recognized as equivalent research contributions to journal articles.

Articles and proceedings

1. Dead Zone of Accountability: Why Social Claims in Machine Learning Research Should Be Articulated and Defended.
Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2025, forthcoming.
[\[Paper\]](#) or [\[Proceeding Poster\]](#)
Tianqi Kou, Dana Calacci, and Cindy Lin
2. From Model Performance to Claim: How a Change of Focus in Machine Learning Replicability Can Help Bridge the Responsibility Gap.
Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2024, Pages 1002 – 1013. [\[Paper\]](#)
Tianqi Kou

Under submission

3. From Managing Claims to Managing Uncertainty: A Relational Account of AI Hype
Big Data & Society, 2026
Tianqi Kou, Dana Calacci, Nasser Eledroos, David Gray Widder, Dasha Pruss

Preprints

4. A Quantitative Machine Learning Approach to Master's Students' Admission for Professional Institutions.

[Preprint available upon request]

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TALKS / PANELS / WORKSHOPS/ SYMPOSIUMS

1. What is the Hype: A Relational Conception of Hype in AI-based Tools in Policing Context.

Workshop Presentation. Hype Studies Conference; Barcelona, Spain. Sept. 2025.

Tianqi Kou, Dana Calacci, Nasser Eledroos, David Gray Widder, Dasha Pruss

2. Claim Replicability and the Responsibility Gap.

Invited Talk, Digital Life Initiative, Cornell Tech, NYC. Oct. 2024.

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3. A Feminist Conception of Replicability for Machine Learning Research

Poster. Institute for Computational and Data Sciences Symposium. Oct. 2024.

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4. The Function of Replication Studies in Machine Learning Research.

Workshop Presentation. Philosophy of Science Meets Machine Learning; Tübingen, Germany. Sept. 2025.

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5. The Underestimation and Overestimation of Reproducibility in Machine Learning Research Claims.

Poster. Institute for Computational and Data Sciences Symposium (2023)

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CONFERENCE PARTICIPATION

2025	AAAI/ACM Conference on Artificial Intelligence Ethics and Society, Madrid, Spain. Author Hype Studies Conference, Barcelona, Spain. Author Privacy Law Scholars Conference, Los Angeles, CA.
2024	ACM Conference on Fairness, Accountability, and Transparency. Author Northeast HCI, Carnegie Mellon University, Pittsburgh, PA.
2023	Association for Library and Information Science Education, Pittsburgh, PA.
2022	Workshop on the Reproducibility Crisis in ML-based Science, Princeton University.

TEACHING

Penn State University

AY26-27	IST 451: Network Security / Teaching Assistant
AY25-26	PHIL 102: Bioethics / Invited lecturer
AY23-24	IST 402: Emerging Issues and Technologies / Teaching Assistant

AY23-24 DS 200: Intro to Data Science / Teaching Assistant
AY22-24 IST 230: Language, Logic, and Discrete Mathematics (3 semesters) / Teaching Assistant
AY21-22 DS 310: Machine Learning / Teaching Assistant

Fordham University

AY17-18 CIS S5800: Machine Learning (2 semesters) / Teaching Assistant
AY18-19 SDGB 7844: Statistical Methods and Computation II / Teaching Assistant

Harbin Institute of Technology, eHealth Research Institute

AY14-15 Intro to Python and R for Digital Health / Instructor Assistant

SERVICE

2026 Program Committee / ACM Conference on Fairness Accountability and Transparency
Reviewer / Big Data & Society
2025 Reviewer / CHI Conference on Human Factors in Computing Systems
Student Track Mentor / AAAI/ACM Conference on AI, Ethics, and Society
Panel Chair / Hype Studies Conference
Reviewer / AAAI/ACM Conference on AI, Ethics, and Society
Program Committee / ACM Conference on Fairness Accountability and Transparency
Reviewer / Big Data & Society
2024 Reviewer / ACM Conference on Human Factors in Computing Systems
2024 Panelist / Queer Graduate Student in STEM, Penn State College of Engineering.